

20260922 Quiz

for (MATH301) Modern Algebra I

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Exercise 1.1.21. Let G be a finite group and let x be an element of G of order n . Prove that if n is odd, then

$$x = (x^2)^k$$

for some k .

Solution. Since n is odd, $k = (n + 1)/2$ is an integer. As $|x| = n$,

$$(x^2)^k = x^{n+1} = x^n x = x.$$

□

Exercise 1.1.25. Prove that if $x^2 = 1$ for all $x \in G$, then G is abelian.

Solution. For every $x \in G$, the hypothesis gives $x^{-1} = x$. Hence for arbitrary $a, b \in G$,

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba.$$

Thus G is abelian.

□

Exercise 1.1.31. Prove that any finite group G of even order contains an element of order 2. Let

$$t(G) = \{g \in G \mid g \neq g^{-1}\}.$$

Show that $t(G)$ has an even number of elements and every nonidentity element of $G - t(G)$ has order 2.

Solution. If $g \in t(G)$, then $g^{-1} \in t(G)$ and $g \neq g^{-1}$. Thus the elements of $t(G)$ occur in disjoint pairs $\{g, g^{-1}\}$, so $|t(G)|$ is even. If $x \in G - t(G)$, then $x = x^{-1}$; hence every nonidentity element of $G - t(G)$ satisfies $x^2 = 1$ and has order 2. Since $|G|$ and $|t(G)|$ are even, $|G - t(G)|$ is even. As $1 \in G - t(G)$, this set contains at least one nonidentity element, and therefore G contains an element of order 2.

□

Exercise 1.1.34. If x is an element of infinite order in G , prove that the elements x^n , $n \in \mathbb{Z}$, are all distinct.

Solution. Suppose $x^m = x^n$ for some $m, n \in \mathbb{Z}$. If $m \neq n$, assume without loss of generality that $m > n$. Then

$$x^{m-n} = 1,$$

with $m - n > 0$, contradicting that x has infinite order. Hence $m = n$, so the powers x^n are all distinct. □

Exercise 1.2.3. Use the presentation

$$D_n = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$$

to show that every element of D_n which is not a power of r has order 2. Deduce that D_n is generated by the two elements s and sr , both of which have order 2.

Notation. Following MATH000, D_n denotes the dihedral group of symmetries of a regular n -gon, so $|D_n| = 2n$. Dummit–Foote denotes this group by D_{2n} .

Solution. Every element of D_n which is not a power of r has the form sr^i . From $rs = sr^{-1}$ we obtain $r^i s = sr^{-i}$, so

$$(sr^i)^2 = sr^i sr^i = s(sr^{-i})r^i = 1.$$

Since $sr^i \neq 1$, it has order 2. In particular, s and sr have order 2. Also

$$s(sr) = r,$$

so s and sr generate both r and s , hence generate D_n . □

Exercise 1.2.6. Let x and y be elements of order 2 in any group G . Prove that if $t = xy$, then

$$tx = xt^{-1}.$$

Thus, if $n = |xy| < \infty$, then x and t satisfy the same relations in G as s and r do in D_n .

Solution. Since $x^{-1} = x$ and $y^{-1} = y$,

$$t^{-1} = (xy)^{-1} = yx.$$

Hence

$$tx = (xy)x = xyx = x(yx) = xt^{-1}.$$

If $n = |t| < \infty$, then

$$t^n = x^2 = 1, \quad tx = xt^{-1},$$

which are the defining relations for r and s in D_n under $r \leftrightarrow t$ and $s \leftrightarrow x$. □

Exercise 1.2.7. Show that

$$\langle a, b \mid a^2 = b^2 = (ab)^n = 1 \rangle$$

gives a presentation for D_n in terms of the two generators $a = s$ and $b = sr$ of order 2. Show that the relations for r and s follow from the relations for a and b , and conversely, that the relations for a and b follow from those for r and s .

Solution. Set $a = s$ and $b = sr$. Then

$$a^2 = s^2 = 1, \quad b^2 = (sr)^2 = 1, \quad ab = s(sr) = r,$$

so $(ab)^n = r^n = 1$.

Conversely, assume $a^2 = b^2 = (ab)^n = 1$ and set

$$s = a, \quad r = ab.$$

Then $s^2 = 1$ and $r^n = 1$. Since $a^{-1} = a$ and $b^{-1} = b$,

$$r^{-1} = (ab)^{-1} = ba,$$

and therefore

$$rs = (ab)a = aba = a(ba) = sr^{-1}.$$

Thus the two sets of generators and relations determine one another, giving the stated presentation of D_n . □

Exercise 1.2.15. Find a set of generators and relations for $\mathbb{Z}/n\mathbb{Z}$.

Solution. A generating set is $\{[1]_n\}$, and the defining relation is

$$n[1]_n = [0]_n.$$

Equivalently, in multiplicative presentation notation,

$$\mathbb{Z}/n\mathbb{Z} \cong \langle x \mid x^n = 1 \rangle.$$

□

Exercise 1.3.5. Find the order of

$$(1 \ 12 \ 8 \ 10 \ 4)(2 \ 13)(5 \ 11 \ 7)(6 \ 9).$$

Solution. The displayed cycles are disjoint and have lengths 5, 2, 3, 2. Hence the order is

$$\text{lcm}(5, 2, 3, 2) = 30.$$

□

Exercise 1.3.13. Show that an element has order 2 in S_n if and only if its cycle decomposition is a product of commuting 2-cycles.

Solution. Let $\sigma \in S_n$ have order 2. Then $\sigma^2 = 1$. If a nontrivial cycle in the disjoint cycle decomposition of σ had length $m \geq 3$, applying σ twice would not fix the elements in that cycle. Hence every nontrivial cycle has length 2. Since disjoint cycles commute, σ is a product of commuting 2-cycles.

Conversely, if σ is a nonempty product of disjoint 2-cycles, then the cycles commute and each has square 1, so $\sigma^2 = 1$. Since $\sigma \neq 1$, it follows that $|\sigma| = 2$. □

Exercise 1.3.14. Let p be a prime. Show that an element has order p in S_n if and only if its cycle decomposition is a product of commuting p -cycles. Show by an explicit example that this need not be the case if p is not prime.

Solution. Suppose $|\sigma| = p$. Then $\sigma^p = 1$. If a nontrivial cycle in the disjoint cycle decomposition of σ has length m , then an m -cycle returns to the identity after p applications only when $m \mid p$. Since p is prime and $m > 1$, we have $m = p$. Thus σ is a product of disjoint, hence commuting, p -cycles.

Conversely, a nonempty product of disjoint p -cycles has p th power equal to 1, while no positive power smaller than p is the identity. Hence its order is p .

The primality assumption is necessary. For example,

$$\sigma = (1\ 2)(3\ 4\ 5\ 6)$$

has order 4, but its cycle decomposition is not a product solely of 4-cycles. □

Exercise 1.3.15. Prove that the order of an element in S_n equals the least common multiple of the lengths of the cycles in its cycle decomposition. Use Exercise 10 and Exercise 24 of Section 1.1.

Solution. Write the disjoint cycle decomposition as

$$\sigma = \sigma_1 \sigma_2 \cdots \sigma_r,$$

where σ_i has length m_i . Since the cycles are disjoint, they commute, and therefore

$$\sigma^k = \sigma_1^k \sigma_2^k \cdots \sigma_r^k.$$

This is the identity if and only if each σ_i^k is the identity, which holds if and only if $m_i \mid k$ for every i . Hence the least positive such k is

$$\text{lcm}(m_1, \dots, m_r).$$

□

Exercise 1.3.18. Find all numbers n such that S_5 contains an element of order n . Use Exercise 15.

Solution. The possible nontrivial disjoint cycle types in S_5 have lengths

$$2, \quad 2+2, \quad 3, \quad 3+2, \quad 4, \quad 5.$$

By Exercise 15 their orders are respectively

$$2, \quad 2, \quad 3, \quad 6, \quad 4, \quad 5.$$

Including the identity, the possible orders are therefore

$$\boxed{1, 2, 3, 4, 5, 6}.$$

□

Exercise 1.6.4. Prove that the multiplicative groups $\mathbb{R} - \{0\}$ and $\mathbb{C} - \{0\}$ are not isomorphic.

Solution. The element $i \in \mathbb{C} - \{0\}$ has order 4. If the two groups were isomorphic, $\mathbb{R} - \{0\}$ would also contain an element of order 4, since isomorphisms preserve element orders. But if $x \in \mathbb{R} - \{0\}$ satisfies $x^4 = 1$, then $x = \pm 1$, so x has order 1 or 2. This is a contradiction. □

Exercise 1.6.14. Let G and H be groups and let $\varphi: G \rightarrow H$ be a homomorphism. Define the kernel of φ to be

$$\ker \varphi = \{g \in G \mid \varphi(g) = 1_H\}.$$

Prove that $\ker \varphi$ is a subgroup of G (cf. Exercise 1.1.26). Prove that φ is injective if and only if $\ker \varphi$ is the identity subgroup of G .

Solution. Since $\varphi(1_G) = 1_H$, we have $1_G \in \ker \varphi$. If $a, b \in \ker \varphi$, then

$$\varphi(ab) = \varphi(a)\varphi(b) = 1_H,$$

and

$$\varphi(a^{-1}) = \varphi(a)^{-1} = 1_H.$$

Thus $\ker \varphi$ is closed under products and inverses, so $\ker \varphi \leq G$.

If φ is injective and $g \in \ker \varphi$, then

$$\varphi(g) = 1_H = \varphi(1_G),$$

so $g = 1_G$. Hence $\ker \varphi = \{1_G\}$. Conversely, suppose $\ker \varphi = \{1_G\}$ and $\varphi(a) = \varphi(b)$. Then

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = 1_H,$$

so $ab^{-1} = 1_G$, and therefore $a = b$. Hence φ is injective. □

Exercise 1.6.19. Let

$$G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{N}\}.$$

Prove that for any fixed integer $k > 1$, the map from G to itself defined by

$$z \mapsto z^k$$

is a surjective homomorphism but is not an isomorphism.

Solution. Define $\varphi(z) = z^k$. Since multiplication in $\mathbb{C}$ is commutative,

$$\varphi(zw) = (zw)^k = z^k w^k = \varphi(z)\varphi(w),$$

so φ is a homomorphism.

Let $w \in G$. Choose $z \in \mathbb{C}$ with $z^k = w$. If $w^n = 1$, then

$$z^{kn} = (z^k)^n = w^n = 1,$$

so $z \in G$ and $\varphi(z) = w$. Thus φ is surjective.

Finally, let $\zeta = e^{2\pi i/k}$. Then $\zeta \neq 1$, $\zeta \in G$, and

$$\varphi(\zeta) = \zeta^k = 1 = \varphi(1).$$

Hence φ is not injective and therefore is not an isomorphism. □

Exercise 1.6.22. Let A be an abelian group and fix some $k \in \mathbb{Z}$. Prove that the map

$$a \mapsto a^k$$

is a homomorphism from A to itself. If $k = -1$, prove that this homomorphism is an isomorphism, i.e., an automorphism of A .

Solution. For $a, b \in A$, the elements commute, so the exponent laws give

$$(ab)^k = a^k b^k.$$

Hence the map $\varphi(a) = a^k$ satisfies

$$\varphi(ab) = \varphi(a)\varphi(b),$$

and is a homomorphism.

If $k = -1$, then $\varphi(a) = a^{-1}$ and

$$\varphi(\varphi(a)) = (a^{-1})^{-1} = a.$$

Thus $\varphi^{-1} = \varphi$, so φ is an automorphism of A . □

Exercise 1.6.25. Let $n \in \mathbb{N}$, let r and s be the usual generators of D_n , and let $\theta = 2\pi/n$.

(a) Prove that

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

is the matrix of the linear transformation which rotates the x, y plane about the origin in a counterclockwise direction by θ radians.

(b) Prove that the map $\varphi: D_n \rightarrow \text{GL}_2(\mathbb{R})$ defined on generators by

$$\varphi(r) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \varphi(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism of D_n into $\text{GL}_2(\mathbb{R})$.

(c) Prove that the homomorphism φ in part (b) is injective.

Solution. Denote the matrix in part (a) by $\mathbf{R}$ and the matrix assigned to s in part (b) by $\mathbf{S}$.

(a) A counterclockwise rotation by θ sends

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}.$$

These images are the columns of $\mathbf{R}$, proving part (a).

(b) Since $\mathbf{R}$ is rotation by $2\pi/n$, $\mathbf{R}^n = \mathbf{I}$, and directly $\mathbf{S}^2 = \mathbf{I}$. Moreover,

$$\mathbf{RS} = \begin{pmatrix} -\sin \theta & \cos \theta \\ \cos \theta & \sin \theta \end{pmatrix} = \mathbf{SR}^{-1}.$$

Thus $\mathbf{R}$ and $\mathbf{S}$ satisfy $r^n = s^2 = 1$ and $rs = sr^{-1}$. Hence $r \mapsto \mathbf{R}$, $s \mapsto \mathbf{S}$ extends to a homomorphism $\varphi: D_n \rightarrow \text{GL}_2(\mathbb{R})$.

(c) By Exercise 1.6.14, it suffices to show $\ker \varphi = \{1\}$. Every element of D_n is r^i or sr^i with $0 \leq i < n$. If $\mathbf{R}^i = \mathbf{I}$, then rotation by $2\pi i/n$ is the identity, so $i = 0$. Also

$$\det(\mathbf{SR}^i) = \det(\mathbf{S}) \det(\mathbf{R}^i) = -1 \neq \det(\mathbf{I}),$$

so $\mathbf{SR}^i \neq \mathbf{I}$ for every i . Hence $\ker \varphi = \{1\}$, and φ is injective. □